

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03/08/2026	Duration:	60 minutes

Code of Conduct

I Juhi Surin (192525148) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

- 1. Study and Analyze the Scenario**
 - o Carefully understand the given problem statement.
 - o Identify the business domain, stakeholders, existing challenges, and system objectives.
- 2. Identify Functional and Non-Functional Requirements**
 - o List all functional requirements required by the proposed system.
 - o Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
- 3. Identify Actors and Use Cases**
 - o Identify all primary and secondary actors interacting with the system.
 - o List the major use cases associated with each actor.
 - o Draw a Use Case Diagram representing the interactions between actors and the system.
- 4. Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

 - o Introduction
 - o Problem Description

- o Scope of the System
- o Functional Requirements
- o Non-Functional Requirements
- o Use Case Descriptions
- o Data Requirements
- o External Interface Requirements
- o System Features

5. Identify Assumptions and Constraints

- o List the assumptions considered while developing the system.
- o Identify technical, operational, economic, legal, and time-related constraints affecting the project.

6. Validate Requirement Completeness and Consistency

- o Verify whether all stakeholder requirements have been addressed.
- o Check for ambiguity, redundancy, inconsistency, and missing requirements.
- o Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS)	SRS is well-structured, complete, professionally organized, and	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or

Documentation	includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/5

SAVEETHA SCHOOL OF ENGINEERING, SIMATS

Department of Computer Science and Engineering

SCENARIO-BASED REQUIREMENT ANALYSIS

Software Requirement Specification (SRS)

CO2 – Assessment Tool 1

Problem Statement #37

AI-Based Digital Waste Recycling Marketplace and Resource Recovery System

Submitted by: Juhi Surin

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Course: CSA10 – Software Engineering

Under the supervision of: Latha R

CO Assessed: CO2 — Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills

Date of Submission: 03 August 2026

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1. Introduction

1.1 Purpose

This document is a Software Requirement Specification (SRS) prepared in response to Problem Statement #37 — the AI-Based Digital Waste Recycling Marketplace and Resource Recovery System. It performs a scenario-based requirement analysis of the assigned problem: understanding the business context, identifying the stakeholders and their goals, deriving functional and non-functional requirements, modelling the actors and use cases, specifying the data and interface requirements, and validating the resulting requirement set for completeness and consistency. The document is intended to give a design and development team a single, unambiguous reference for what the system must do before architecture and sprint planning begin.

1.2 Document Conventions

Functional requirements are labelled FR-<module>-<number> (e.g., FR-REG-1), non-functional requirements NFR-<number>, use cases UC-<number>, and data entities DE-<number>, so that every requirement can be traced back to its source objective in Section 11. Priority is expressed using MoSCoW (Must / Should / Could / Won't) unless a table specifies otherwise. “The system shall” is used for mandatory functional requirements, consistent with IEEE 830 SRS conventions.

1.3 Intended Audience and Reading Suggestions

This SRS is written for four audiences who will each read it differently: the development team (Sections 4–9, for build-level detail), the course evaluator / supervisor (Sections 2, 6 and 11, for analytical depth), project stakeholders such as municipal or recycling-industry partners (Sections 2 and 3, for scope and outcomes), and quality assurance (Sections 4, 5 and 6, since acceptance testing is derived directly from these).

1.4 Project Scope (Summary)

The system is a cloud-hosted digital marketplace and AI classification platform that connects household and industrial waste generators, waste collection agencies, recycling facilities, and buyers of recycled material, while giving regulators and sustainability officers verified reporting. A detailed in-scope / out-of-scope breakdown is given in Section 3.

1.5 References

Base problem statement: “37. AI-Based Digital Waste Recycling Marketplace and Resource Recovery System” (assignment problem bank). Prior submissions for this project — the Case Study Analysis Report (CO1-AT1) and the Agile Sprint Planning & User Story Workshop (CO1-AT2) — are used as supporting analytical background and are referenced throughout this document. IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications, is used as the structural reference for this SRS.

2. Problem Description

2.1 Business Domain

The domain is municipal and industrial solid-waste management within the broader circular-economy and environmental-technology sector. Rapid urbanisation has increased the volume of recyclable waste — plastics, paper, metal, e-waste, and glass — generated by households and industries, while the channels for recovering that material remain informal, manual, and disconnected. Governments and environmental organisations are actively promoting circular-economy practices, and AI, IoT, cloud computing, and digital-marketplace technology are now mature enough to coordinate waste collection, segregation, trading, and recycling at scale.

2.2 Problem Statement

Existing waste-management systems lack effective coordination between waste generators, recycling facilities, collection agencies, and buyers of recyclable material. Households have no reliable way to know what is recyclable or who will take it; industries negotiate scrap prices informally, transaction by transaction, with whichever trader is available; collection agencies plan pickup routes manually with no visibility into upcoming volumes; and recyclers receive unsorted, inconsistent material because there is no shared classification standard. The absence of a common data layer means recycling rates stay low and landfill dependency stays high, even in places where recyclable material is genuinely available — the system does not lack recyclable waste, it lacks coordination.

2.3 System Objectives

The proposed AI-powered digital marketplace must, in order of operational dependency:

- Digitize recyclable-waste trading, replacing informal, price-opaque scrap transactions.
- Automatically classify waste materials using AI (image- or description-based) to reduce manual sorting errors.
- Connect waste generators, collection agencies, and recyclers/buyers inside one coordinated workflow.
- Optimize collection schedules and pickup routes to reduce travel time and fuel cost.
- Continuously monitor recycling activity so the platform and regulators have a live, auditable picture.
- Generate sustainability analytics — recycling rate, landfill diversion, emissions saved.
- Promote circular-economy practices by making reuse and resale of materials the easier option.
- Reduce the overall volume of waste reaching landfills.

2.4 Key Stakeholders

Nine stakeholder groups interact with the system, each with a different technical comfort level and a different primary interest — a design constraint that shapes the requirements in Sections 4–8.

Stakeholder	Role / Responsibility	Primary Interest / Pain Point Today
Household Waste Generators	List household recyclables (plastic, paper, e-waste); request pickup.	No easy way to know what is recyclable or who buys it; low price transparency.
Industrial / Commercial Generators	Generate bulk, often specialised recyclable waste streams.	Inconsistent scrap pricing; compliance reporting burden; unreliable pickup timing.
Waste Collection Agencies	Pick up listed waste and transport it to the correct recycler.	Manually-planned, inefficient routes; no visibility into upcoming volumes.
Recycling Facilities / Recyclers	Process recyclable material sourced through the marketplace.	Inconsistent supply and quality; material arrives unsorted or contaminated.
Buyers of Recyclable Materials	Purchase processed, sorted recycled raw material.	Difficulty sourcing verified, consistent-quality material at scale.
Platform Administrators	Operate the marketplace; moderate listings; ensure trust.	Preventing fraud/mis-classification; keeping the AI model accurate over time.
Environmental Regulatory Bodies	Set policy; audit recycling and landfill statistics.	Lack of real-time, reliable data on actual recycling activity.
Sustainability Officers (Municipal/CSR)	Track circular-economy KPIs for a city or organisation.	No reliable dashboards or impact reporting today.
AI/ML & Platform Development Team	Build and maintain the classification model and platform.	Needs clear requirements and clean, labelled data pipelines.

Table 2.1 — Key stakeholder roles and primary interests

2.5 Current-Approach Gap Analysis

Today, households hand recyclables to municipal trucks (which rarely separate at pickup) or sell to door-to-door scrap collectors who price by rough visual estimation; industries deal informally with a handful of scrap dealers; and recyclers buy from dealers in bulk with no visibility into origin or handling. This works well enough for low-friction, low-accountability transactions, but it produces five structural gaps that the proposed system must close:

- No standard material classification — waste is sorted by eye, so contamination and misclassification are common.
- No price transparency — sellers have no reference point, so pricing is inconsistent.
- No route or schedule optimisation — agencies plan pickups manually, wasting fuel and time.
- No data trail — there is no reliable record of material recycled versus landfilled.
- Fragmented stakeholder communication — generators, collectors, and recyclers rarely interact directly, so supply and demand are never efficiently matched.

3. Scope of the System

3.1 In Scope

- Web and mobile application for households, industries, collection agencies, and recyclers.
- AI-based waste classification module (image- and description-based material identification).
- Marketplace and matching logic connecting sellers of recyclable material with buyers/recyclers.
- Route and pickup-schedule optimisation for collection logistics.
- Analytics dashboard for recycling rates and sustainability metrics.
- Integration hooks for IoT-enabled smart bins (fill-level sensors) for pickup triggers.
- Secure payments and transaction settlement between sellers and buyers.
- Role-based registration, authentication, and account management for all stakeholder types.

3.2 Out of Scope

- Manufacturing or physical design of smart bins and IoT hardware.
- Operation of the recycling plants themselves (mechanical/chemical recycling processes).
- Government policy-making — the platform only reports data that regulators can act on.
- Physical transport fleet ownership — vehicles and drivers remain the responsibility of collection-agency partners.

4. Functional Requirements

Functional requirements are grouped by module and prioritised using MoSCoW. Together they cover the eight system objectives in Section 2.3; the traceability from objective to requirement is validated in Section 11.1.

4.1 User & Account Management

ID	Requirement	Priority
FR-REG-1	The system shall allow a new user to register and select a role (household, industry, recycler, agency, buyer, admin).	Must
FR-REG-2	The system shall verify a new account via OTP or e-mail before activation.	Must
FR-REG-3	The system shall route each user to a role-specific dashboard after login.	Must
FR-REG-4	The system shall allow users to manage their profile, address, and contact details.	Should

4.2 Waste Listing & AI Classification

ID	Requirement	Priority
FR-LST-1	The system shall allow a generator to list recyclable waste with photo(s), quantity, and description.	Must
FR-LST-2	The system shall allow an industrial user to post bulk waste lots with material grade and preferred trading price.	Must
FR-CLS-1	The system shall automatically classify an uploaded waste photo into a material category (plastic, e-waste, metal, glass, organic, paper) with a confidence score.	Must
FR-CLS-2	The system shall flag a listing “Needs Manual Review” whenever AI confidence is below 60%.	Must
FR-CLS-3	The system shall flag contaminated or mixed-material listings and notify the recycler before pickup.	Should

4.3 Marketplace, Matching & Discovery

ID	Requirement	Priority
FR-MKT-1	The system shall allow recyclers/buyers to search and filter listings by material type, quantity, and location radius.	Must
FR-MKT-2	The system shall recommend the nearest suitable recycling centres for a classified listing, ranked by distance and material acceptance.	Must
FR-MKT-3	The system shall notify a recycler within 2 minutes when a new listing matches their registered material preferences and service radius.	Should
FR-MKT-4	The system shall allow a buyer to place an order and complete a secure payment for a listing.	Must

4.4 Collection Scheduling & Logistics

ID	Requirement	Priority
FR-SCH-1	The system shall allow a household/industrial user to book or reschedule a pickup slot.	Must
FR-SCH-2	The system shall generate an optimised pickup route and schedule for a collection agency handling multiple listings.	Should
FR-SCH-3	The system shall ingest IoT smart-bin fill-level data to trigger a pickup request automatically.	Could

4.5 Monitoring, Compliance & Analytics

ID	Requirement	Priority
FR-MON-1	The system shall track the status of every waste item (Listed → Matched →	Should

ID	Requirement	Priority
	Collected → Recycled) end to end.	
FR-MON-2	The system shall provide regulators an exportable, verified report of recycled quantities per recycler and date range.	Could
FR-ANL-1	The system shall display recycling rate, estimated emissions avoided, and landfill-diversion percentage on a sustainability dashboard, filterable by date and region.	Should
FR-ANL-2	The system shall generate a monthly PDF sustainability summary automatically for each recycler.	Could
FR-ENG-1	The system shall credit eco-points to a household account on completion of a recycling transaction.	Could

5. Non-Functional Requirements

Non-functional requirements define the quality attributes the platform must satisfy regardless of which feature is being used. They are grouped using the standard ISO 25010-aligned categories requested by the assignment brief.

ID	Category	Requirement
NFR-1	Performance	AI waste classification shall return a material category and confidence score within 3–5 seconds for a typical image on a standard mobile connection.
NFR-2	Scalability	The platform shall support a growing number of concurrent users, listings, and cities without architectural redesign, via horizontally-scalable cloud services.
NFR-3	Reliability / Availability	Core marketplace and matching services shall maintain at least 99.5% uptime, with automated monitoring and redundant infrastructure.
NFR-4	Usability	The interface shall be usable by a first-time household user with no technical background, requiring no more than three steps to list an item.
NFR-5	Security	User data, location details, and payment information shall be encrypted in transit and at rest; access shall be role-based.
NFR-6	Maintainability	Modules (classification, matching, routing, analytics) shall be independently deployable so the AI model or matching logic can be updated without downtime elsewhere.
NFR-7	Interoperability	The platform shall expose documented APIs for maps, payment gateways, and IoT sensor ingestion so third-party services can be swapped without redesign.
NFR-8	Auditability	Every status change and transaction shall be timestamped and immutable once recorded, to support regulatory audit trails.
NFR-9	Accessibility	Web and mobile interfaces shall meet WCAG 2.1 AA colour-contrast and screen-reader labelling for core listing and search flows.

6. Actors and Use Cases

6.1 Primary Actors

Primary actors initiate interactions with the system to accomplish a goal:

- Household Waste Generator — lists small-quantity recyclables and books pickups.
- Industrial / Commercial Waste Generator — posts bulk, graded waste lots.
- Waste Collection Agency — executes pickups and consumes optimised routes.
- Recycler / Recycling Facility — discovers, claims, and processes listed material.
- Buyer of Recyclable Material — purchases processed recycled raw material.
- Platform Administrator — moderates listings and manages platform operations.

6.2 Secondary / Supporting Actors

- Regulatory / Municipal Body — consumes compliance reports; does not directly operate the marketplace.
- AI Classification Engine — a system actor invoked automatically whenever a listing is created.
- Payment Gateway (external system) — invoked during checkout to settle transactions.
- Maps / Geolocation Service (external system) — invoked for recommendation and route optimisation.
- IoT Smart-Bin Sensors (external system) — push fill-level events that can trigger a pickup request.

6.3 Use Case Summary by Actor

Actor	Associated Use Cases
Household Waste Generator	Register & Verify Account; List Recyclable Waste; Book/Reschedule Pickup; Earn Eco-points; Receive Notifications
Industrial Waste Generator	Register & Verify Account; List Recyclable Waste (bulk); View Sustainability Dashboard
Waste Collection Agency	Optimize Collection Route; Book/Reschedule Pickup; Track Waste Status
Recycler / Recycling Facility	Browse & Filter Listings; View Recommended Recycling Centres; Track Waste Status; View Sustainability Dashboard
Buyer of Recyclable Material	Browse & Filter Listings; Place Order & Make Payment
Platform Administrator	Moderate Listings & Manage Platform; Register & Verify Account (approvals); Track Waste Status
Regulatory / Municipal Body	Generate Compliance Report; View Sustainability Dashboard

6.4 Use Case Diagram

The diagram below models all primary and secondary actors against the system boundary. Dashed <<include>> arrows mark a use case that always invokes another (e.g., listing waste always includes AI classification); the dashed <<extend>> arrow marks optional behaviour (notification on order placement).

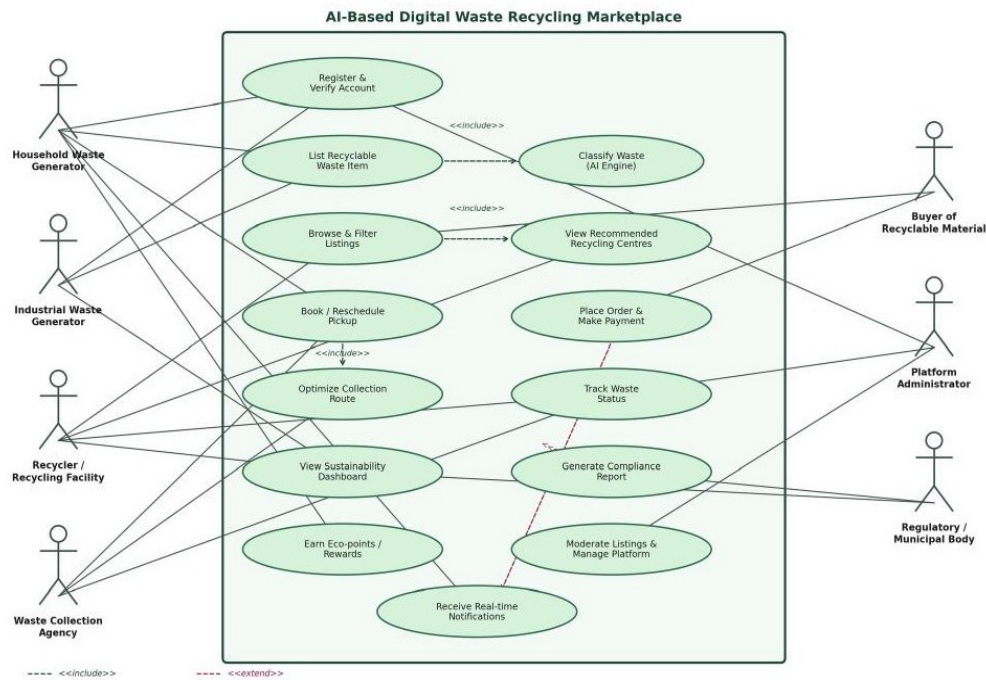


Figure 6.1 — UML Use Case Diagram: AI-Based Digital Waste Recycling Marketplace

6.5 Detailed Use Case Descriptions (UC-1 to UC-5)

The five highest-priority use cases — corresponding to the Sprint 1 minimum viable product identified in the earlier Agile Sprint Planning submission — are described below in full UML use-case format. Remaining use cases follow the same template and are summarised in Section 6.3.

UC-1: Register & Verify Account

Actors	Household / Industrial Generator, Recycler, Collection Agency, Buyer (primary); Platform Administrator (secondary)
Description	A new user creates a role-based account and verifies it before gaining access to role-specific features.
Preconditions	The user has a valid e-mail address or mobile number and is not already registered.
Main Flow	1) User selects a role. 2) User enters profile details. 3) System sends an OTP/e-mail verification code. 4) User submits the code. 5) System creates the account and routes the user to their role dashboard.
Alternate Flow	A3a) If the OTP expires or is entered incorrectly three times, the system locks the attempt for 10 minutes and offers a resend option.

Postconditions	A verified, role-specific account exists and the user can access the marketplace.
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UC-2: List Recyclable Waste Item (includes AI Classification)

Actors	Household / Industrial Generator (primary); AI Classification Engine (included)
Description	A generator uploads a photo and details of waste; the system automatically classifies the material before publishing the listing.
Preconditions	The user holds a verified account.
Main Flow	1) User uploads at least one photo, quantity, and description. 2) System invokes the AI Classification Engine. 3) Engine returns a material category and confidence score. 4) If confidence $\geq 60\%$, the listing is published as "Pending Classification → Classified". 5) System notifies matching recyclers per FR-MKT-3.
Alternate Flow	A3a) If confidence $< 60\%$, the listing is flagged "Needs Manual Review" instead of auto-publishing (FR-CLS-2). A3b) If the photo shows mixed/contaminated material, the listing is flagged "Contamination Risk" and the recycler is notified before pickup (FR-CLS-3).
Postconditions	A classified, searchable listing exists on the marketplace.

UC-3: Browse & Filter Listings

Actors	Recycler, Buyer of Recyclable Material
Description	A recycler or buyer searches the marketplace by material type, quantity, and location to source raw material.
Preconditions	At least one classified listing exists.
Main Flow	1) User opens Search. 2) User filters by material type, location radius, and quantity. 3) System returns matching, currently-available listings sorted by distance. 4) User selects a listing to view full detail, including recommended recycling centres (includes UC-4).
Alternate Flow	A3a) If no listings match, the system suggests widening the radius or material type.
Postconditions	The user has identified one or more listings to pursue.

UC-4: View Recommended Recycling Centres

Actors	Household / Industrial Generator; Recycler
Description	After a listing is classified, the system recommends the nearest recycling centres capable of accepting that material.
Preconditions	The listing has been successfully classified.
Main Flow	1) User opens the listing detail page. 2) System queries the geolocation service for nearby centres accepting the classified material. 3) The three nearest suitable centres are shown, ranked by distance and acceptance rate.

Alternate Flow	A2a) If no centre within the default radius accepts the material, the system expands the radius and re-queries.
Postconditions	The generator can choose a recycling destination for the listed item.

UC-5: Optimize Collection Route

Actors	Waste Collection Agency
Description	An agency requests an optimised sequence of stops for multiple pickups scheduled on the same day.
Preconditions	At least two pickups are confirmed for the agency on the requested day.
Main Flow	1) Agency opens Route Planner. 2) Agency selects the date/list of confirmed pickups. 3) System calls the route-optimisation engine. 4) System returns a sequence of stops minimising total travel distance and time.
Alternate Flow	A3a) If a stop becomes unavailable (e.g., pickup cancelled) after the route is generated, the system re-optimises automatically.
Postconditions	The agency has an actionable, time-sequenced pickup route.

7. Data Requirements

7.1 Key Data Entities

The platform is built around seven core data entities. Their relationships mirror the process flow: a User creates a WasteListing, which the ClassificationResult enriches, which a Recycler/Buyer converts into a Transaction, which a Pickup fulfils, and whose outcome feeds the SustainabilityRecord.

Entity	Key Attributes	Notes
User	user_id, role, name, verified_flag, address, contact	One record per stakeholder account; role drives dashboard and permissions.
WasteListing	listing_id, owner_id, photo_url(s), quantity, description, status, price	status moves Listed → Matched → Collected → Recycled.
ClassificationResult	result_id, listing_id, material_category, confidence_score, contamination_flag	Produced by the AI Classification Engine; drives auto-publish vs. manual review.
Transaction	txn_id, listing_id, buyer_id, amount, payment_status, timestamp	Created on checkout; links to the external payment gateway.
Pickup	pickup_id, listing_id, agency_id, scheduled_slot, route_id, status	Created when a schedule is booked; consumed by the route optimiser.
SustainabilityRecord	record_id, region, recycler_id,	Aggregated periodically to power the analytics

Entity	Key Attributes	Notes
	quantity_recycled, emissions_avoided, period	dashboard and monthly reports.
ComplianceReport	report_id, recycler_id, date_range, verified_quantity, export_format	Generated on demand for regulators; sourced from SustainabilityRecord and Transaction.

7.2 Data Flow and Retention

Waste photographs and classification results flow from the mobile/web client to the AI Classification Engine and back to the WasteListing record; transaction and payment data flow through the payment gateway and are stored only in tokenised form. Sustainability and compliance data are derived (not manually entered) so that reporting is a byproduct of normal marketplace use rather than a separate task. Personally identifiable data (address, contact) is retained only as long as the account is active plus a regulatory-mandated audit period; transaction records are retained longer to satisfy compliance requirements described in Section 10.2.

8. External Interface Requirements

8.1 User Interfaces

- A responsive web portal and a native/hybrid mobile app share a single role-based interface, so a first-time household user and an industrial operations manager see the same platform with role-appropriate views.
- Waste listing uses a guided, photo-first form (camera capture or gallery upload) to keep the entry barrier low for non-technical users.
- Dashboards (search, tracking, sustainability) use consistent card- and chart-based layouts across web and mobile.

8.2 Hardware Interfaces

- Device camera / photo library, for waste-listing image capture.
- GPS / location services, for proximity-based search and recommendation.
- IoT smart-bin fill-level sensors, which push events over a lightweight messaging protocol (e.g., MQTT) to trigger automatic pickup requests (optional module, FR-SCH-3).

8.3 Software Interfaces

Interface	Purpose
AI/ML Classification Service	Computer-vision model exposed as an internal microservice API; receives an image, returns category + confidence score.
Maps / Geolocation API	Distance calculation for recommended recycling centres and route optimisation.
Payment Gateway	Secure checkout and settlement between buyers and sellers of recyclable material.

Interface	Purpose
Cloud Object Storage	Stores listing photographs and generated PDF sustainability/compliance reports.
Notification Service (push / SMS / email)	Delivers match, pickup, and status-change alerts to relevant users in real time.
Cloud Database	Persists all entities described in Section 7.1 with role-based access control.

8.4 Communication Interfaces

- All client-server communication uses HTTPS/TLS 1.2+.
- Real-time notifications use a push-notification service for mobile and WebSocket/SSE for the web dashboard.
- IoT sensor ingestion uses MQTT over TLS, bridged into the platform's event pipeline.

9. System Features

System features aggregate related functional requirements into deliverable, demo-able capabilities, ordered by the priority already established in Section 4 and validated against stakeholder value in Section 11.

Feature	Priority	Key Requirements Satisfied
Role-based Registration & Account Management	Must	FR-REG-1 to FR-REG-4
AI-Powered Waste Listing & Classification	Must	FR-LST-1, FR-LST-2, FR-CLS-1 to FR-CLS-3
Marketplace Search, Matching & Recommendation	Must	FR-MKT-1 to FR-MKT-4
Pickup Scheduling & Route Optimisation	Should	FR-SCH-1 to FR-SCH-3
Recycling Activity Monitoring & Compliance	Should / Could	FR-MON-1, FR-MON-2
Sustainability Analytics & Reporting	Should / Could	FR-ANL-1, FR-ANL-2
Community Engagement (Eco-points)	Could	FR-ENG-1

10. Assumptions and Constraints

10.1 Assumptions

- Generators have access to a smartphone or camera-enabled device capable of taking a usable photo of the waste item.
- A sufficient, representative labelled image dataset can be assembled to train an initial classification model above the 60% confidence threshold used for auto-publishing.
- Collection agencies and recyclers are willing to adopt a digital workflow in place of, or alongside, their existing informal channels.
- Basic internet connectivity is available in the target service area, even if intermittent in some regions.
- A third-party payment gateway and maps/geolocation API are available for integration and can be contracted within the project budget.
- Regulatory bodies will accept digitally-generated, platform-verified compliance reports as valid audit evidence.

10.2 Constraints

Type	Constraint
Technical	The AI classification model must run inference within the 3–5 second performance budget (NFR-1) on standard mobile connections, limiting model size/complexity.
Technical	The system must integrate with heterogeneous, sometimes low-cost IoT smart-bin sensors with limited protocol support.
Operational	Collection agencies retain ownership of physical vehicles and staff; the platform can only recommend, not enforce, routes and timing.
Operational	Onboarding informal-sector participants (e.g., existing kabadiwalas) requires training and incentive design outside pure software scope.
Economic	Cloud AI-inference and storage costs scale with listing volume and photo size, constraining the sprint budget for the classification service.
Legal / Compliance	User data, especially location and payment information, must comply with applicable data-protection and e-waste handling regulations (NFR-5, NFR-8).
Time-related	The assignment's Sprint 1 scope (Section 6.5 use cases) must be deliverable within a single two-week sprint at an assumed team velocity of ~35 story points.

11. Requirement Validation

Requirement validation checks that every stakeholder objective from Section 2 is actually addressed by at least one requirement, and that the requirement set itself is free of obvious ambiguity, redundancy, and inconsistency.

11.1 Completeness — Objective-to-Requirement Traceability

Objective (Section 2.3)	Addressed By
Digitize recyclable waste trading	FR-LST-1, FR-LST-2, FR-MKT-1, FR-MKT-4, UC-2, UC-3
Classify waste materials using AI	FR-CLS-1, FR-CLS-2, FR-CLS-3, UC-2
Connect waste generators with recyclers	FR-MKT-1, FR-MKT-2, FR-MKT-3, UC-3, UC-4
Optimize waste collection schedules	FR-SCH-1, FR-SCH-2, FR-SCH-3, UC-5
Monitor recycling activities	FR-MON-1, FR-MON-2
Generate sustainability analytics	FR-ANL-1, FR-ANL-2, NFR-8
Promote circular-economy practices	FR-ENG-1, FR-ANL-1
Reduce landfill waste	FR-MKT-2, FR-ANL-1 (outcome measured, not directly caused, by software)

Every stated objective maps to at least one functional requirement and use case, and every functional requirement maps back to an objective — no orphan requirements were introduced. This confirms completeness against the problem statement as written.

11.2 Consistency and Ambiguity Review

- Terminology check: “Recycler” and “Recycling Facility” are used interchangeably in the source problem statement; this SRS standardises on “Recycler / Recycling Facility” throughout to remove ambiguity.
- Threshold consistency: the 60% AI-confidence threshold (FR-CLS-2) and the 3–5 second classification-latency budget (NFR-1) are cross-referenced in UC-2 so implementation and testing use the same numbers — no conflicting values exist elsewhere in the document.
- No two functional requirements were found to overlap in scope; FR-MKT-2 (recommendation) and UC-4 were checked against FR-SCH-2 (route optimisation) to confirm they use the geolocation service for different purposes (destination discovery vs. multi-stop sequencing) and are not duplicative.
- Redundancy check: sustainability reporting appears in both FR-ANL-1 (dashboard) and FR-ANL-2 (recycler PDF report); these were kept as separate requirements because they serve different actors (regulators/officers vs. individual recyclers) and different delivery mechanisms, not because of duplication.

11.3 Identified Gaps and Suggested Improvements

- Dispute resolution is implied (Section 6.1, Administrator role) but not yet specified as a functional requirement; a future iteration should add an explicit “Raise/Resolve Listing Dispute” use case.
- Offline or low-connectivity behaviour for the mobile app is not currently specified; given the target user base includes areas with intermittent internet (Section 10.1), a non-functional requirement for offline listing drafts with later sync is recommended.

- Data-retention periods (Section 7.2) are described qualitatively; a follow-up iteration should convert these into explicit, numeric retention-policy requirements once the applicable legal framework is confirmed.
- Accessibility (NFR-9) currently targets only web/mobile listing and search flows; extending WCAG coverage to the analytics dashboard is recommended for the next requirement cycle.

12. Conclusion

This scenario-based requirement analysis confirms that the core barrier to higher recycling rates in the assigned problem statement is coordination, not the absence of recyclable material or willing participants. The requirement set derived here — role-based onboarding, AI-assisted classification, marketplace discovery, logistics optimisation, and sustainability reporting — directly targets each of the five structural gaps identified in Section 2.5, and the traceability check in Section 11.1 confirms every stated system objective is covered by at least one testable requirement. The functional and non-functional requirements, actor/use-case model, and data and interface specifications in this document are sufficiently detailed to hand off directly into architecture design and Agile sprint planning, consistent with the CO2 assessment's focus on requirements analysis, scalable design, and feature prioritisation.